

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element F Initial Template - AT - 3 points

How to use this template

Students: The template below is one way to assess whether your proposed solution is viable and realistic, but the specific assignment given by your instructor may require a different structure or dictate different details. If you make a copy of this template to edit, be sure to delete the instructions.

Element F: Consideration of Design Viability

Judgment about Design Viability (F1)

"Design viability" is the capacity of the proposed solution to solve the problem. Design viability refers to how your design decisions are tethered to your design requirements. For example, if one of your design requirements is that the total cost of the prototype cannot exceed X, then you could present a table of costs for each component of your proposed design solution as justification for why your design is likely to meet that requirement.

"Credible evidence" means it shouldn't just be your hunch that this will work. Be creative about how you validate the viability of your design. You should go beyond just having a stakeholder or expert reviewer say that the design seems viable. For example, you could point to evidence that a sub-system already functions, or you demo a proof of concept, or you've discussed with an expert that said, "I've seen something similar work in this other related situation...", etc.

Use as many rows as needed. Remember, this chart below may help you organize your ideas, but you will need to actually execute the "credible evidence."

Design Viability Description	Connection to Design Requirement from Element C	Credible Evidence?

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<i>Example: The solution is inexpensive</i>	<i>Example: Design does not cost more than \$X</i>	<i>Example: Create a table of costs for each component of the solution.</i>
The repair solution must restore structural integrity	The repair material must have a compressive strength of at least 40 MPa to match the existing concrete	Lab testing repair materials show they met or exceeded 40 MPa strength
The solution should be cost-effective	The total cost should not exceed \$50 per square meter	A cost breakdown of materials and labor demonstrates affordability
The repair must be durable and resistant to weathering	The repair should last at least 10 years without significant deterioration	Case studies and material lifespan data confirm Longevity. Even Though there may be weather delays in any given time we would plaster the repair with film/coat while we work to put on the weather resistance.
The solution should minimize downtime for the repaired structure	The area should be usable within 24 hours of application	Manufacturer specifications or real-world usage confirm curing time. We would be able to obtain a building permit as we would need to give an analyzed site plan.

